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Panos Pardalos
Ilias Kotsireas
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Mathematical Research for Blockchain Economy

1st International Conference MARBLE 2019, Santorini, Greece

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Editors
Panos Pardalos
Department of Industrial and Systems Engineering
University of Florida
Gainesville, FL, USA

Yike Guo
Data Science Institute
Imperial College London
London, UK

Ilias Kotsireas
Wilfrid Laurier University
Waterloo, ON, Canada

William Knottenbelt
Department of Computing
Imperial College London
London, UK

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MARBLE 2019 Conference Proceedings Volume: Preface

This volume presents the proceedings of the 1st International Conference on Mathematical Research for Blockchain Economy (MARBLE), being held in Santorini, Greece from May 6 to 9, 2019. In contrast to most blockchain conferences and forums which are dedicated to business applications, product development or ICO launches, MARBLE focuses on the mathematics and economics behind blockchain, seeking to bridge the gap between practice and theory. It aims to provide a high-profile, cutting-edge platform for mathematicians, computer scientists and economists to present latest advances and innovations in key theories of blockchain.

The call for papers solicited 24 research papers across a number of themes including incentives, governance, topological analysis, cryptoassets and security. Of the submissions, 15 were accepted for publication and presentation, and of the accepted paper, four of the top-ranked submissions have been chosen for consideration for the Best Paper Award, which will be presented in a special session and voted on by conference attendees. The winner will be announced during the conference banquet, which is to be held in the atmospheric setting of the Volcano Blue restaurant.

The technical programme also features keynotes by the following distinguished speakers: Roman Beck, Jihan Wu, Ambre Soubrian, George Giaglis, Patrick McCorry and Garrick Hileman, and tutorials on the subject of smart contracts (Jerome de Tychey) and zero knowledge proofs (Alexandre Pinto). There is also an industry panel focused on the challenges of blockchain-led transformation of economies, which will be chaired by Naeem Aslam.

We thank all authors who submitted their innovative work to MARBLE 2019 this year. In addition, we thank all members of the Technical Programme Committee and other reviewers, everyone who submitted a paper for consideration, the General Chairs, Prof. Yike Guo and Prof. Panos Pardalos, the Organising Chair Kai Sun, the Local Chair Ilias Kotsireas, the Finance Chair Diana O'Malley, the Publicity Chairs Anna Frankowska and Sam Werner, and members of the Centre for Cryptocurrency Research and Engineering who have contributed in many different ways to the organisation effort, particularly Katerina Koutsouri and

Lewis Gudgeon. Finally, we are grateful to our sponsors, the Brevan Howard Centre for Financial Analysis, Asseth and Aventus.io for their generous support.

The Brevan Howard Centre for Financial Analysis at Imperial College Business School would like to thank all those who were involved and participated in the great success of MARBLE 2019 and is looking forward to supporting and being involved with MARBLE 2020 and future events.

Santorini, Greece
May 2019

Panos Pardalos
Ilias Kotsireas
Yike Guo
William Knottenbelt

Contents

Topological Analysis of Bitcoin's Lightning Network

István András Seres, László Gulyás, Dániel A. Nagy and Péter Burcsi

Abstract Bitcoin's Lightning Network (LN) is a scalability solution for Bitcoin allowing transactions to be issued with negligible fees and settled instantly at scale. In order to use LN, funds need to be locked in payment channels on the Bitcoin blockchain (Layer-1) for subsequent use in LN (Layer-2). LN is comprised of many payment channels forming a payment channel network. LN's promise is that relatively few payment channels already enable anyone to efficiently, securely and privately route payments across the whole network. In this paper, we quantify the structural properties of LN and argue that LN's current topological properties can be ameliorated in order to improve the security of LN, enabling it to reach its true potential.

Keywords Bitcoin · Lightning network · Network security · Network topology · Payment channel network · Network robustness

Since its launch, Bitcoin [7] gained a huge popularity due to its publicly verifiable, decentralized, permissionless and censorship-resistant nature. This tremendous popularity and increasing interest in Bitcoin pushed its network's throughput to its limits. Without further advancements, the Bitcoin network can only settle 7 transactions per second (tps), while mainstream centralized payment providers such as Visa and Mastercard can process approximately 40,000 tps in peak hours. Moreover one might need to pay large transaction fees on the Bitcoin network, while also need to wait 6 new blocks ($\sim$1 h) to be published in order to be certain enough that the transaction is included in the blockchain.

I. A. Seres (✉) · L. Gulyás · D. A. Nagy · P. Burcsi
Eötvös Loránd University, Budapest 1053, Hungary
e-mail: istvanseres@caesar.elte.hu

L. Gulyás
e-mail: gulya@hps.elte.hu

D. A. Nagy
e-mail: nagy.da@gmail.com

P. Burcsi
e-mail: bupe@inf.elte.hu

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To alleviate these scalability issues the Lightning Network (LN) was designed in 2016 [8], and launched in 2018, January. The main insight of LN is that transactions can be issued off-blockchain in a trust-minimized manner achieving instant transaction confirmation times with negligible fees, whilst retaining the security of the underlying blockchain. Bidirectional payment channels can be formed on-chain using a construction called Hashed Timelock Contracts (HTLC). Later several payments can take place in a payment channel. The main advantage of payment channels is that one can send and receive thousands of payments with essentially only 2 on-chain transations: the opening and closing channel transactions.

Using these payment channels as building blocks one might establish a payment channel network, where it is not necessary to have direct payment channels between nodes to transact with each other, but they could simply route their payments through other nodes' payment channels. Such a network can be built, because LN achieves payments to be made without any counterparty risk, however efficient and privacy-preserving payment routing remains a challenging algorithmic task [9].

Our contributions. We empirically measure[1] and describe LN's topology and show how robust it is against both random failures and targeted attacks. These findings suggest that LN's topology can be ameliorated in order to achieve its true potential.

1 Background on Lightning Network

In this section we provide a short recap on how LN works. In the following we will use the terms Layer-2 and off-chain interchangeably. LN is a so-called Layer-2 technology, which allows participants issuing transactions without sending a Layer-1 transaction on the Bitcoin parent chain. All parties cooperatively open a channel by locking collateral on the blockchain. The funds can only be released by unanimous agreement or through a pre-defined refund condition [3]. One of the greatest challenge of Layer-2 technologies is to solve how participants can agree on new state updates in a trustless or trust-minimized manner.

Let's take the following toy example: Alice and Bob creates by a single Layer-1 transaction a payment channel with initial balances 10Ƀ and 0Ƀ respectively. Straightaway Alice can issue off-chain transactions to Bob up to 10Ƀ. Let's assume Alice issued 3 off-chain transactions to Bob each worth of 1Ƀ. Afterwards Alice's and Bob's balance should be 7Ƀ and 3Ƀ and neither Alice, nor Bob should be able to redeem previous balances on the parent chain. LN achieves this by a replace by revocation mechanism, namely both parties collectively authorize a new state before revoking the previous state. Upon dispute, the blockchain provides a time period for parties to prove that the published state is a revoked state [3]. Revoking old channel states is achieved by the exchange of revocation secrets. These secrets, hash preimages, are needed to be retained during the channel's lifetime. A penalty mechanism discourages parties from broadcasting older states. If one party broadcasts a revoked

[1] https://github.com/seresistvanandras/LNTopology.

state, the blockchain accepts within a time-window proofs of maleficence from the other party. A successful dispute allots the winning party *all* coins of the channel. In our example if Alice broadcasts a revoked channel state with balances 8Ƀ and 2Ƀ, Bob can prove, that Alice maliciously broadcasted a revoked state. The penalty mechanism grants all the 10Ƀ to Bob.

The great insight of LN, is that if now Alice would like to issue a payment to Cecily, who eventually has already established a payment channel to Bob, then Alice does not need to open a payment channel and create a costly on-chain transaction, rather she can route her payment through her payment channel with Bob to Cecily. However the maximum amount of bitcoins Alice can send to Cecily is the minimum of all the individual balances on the payment route from Alice to Cecily. Hashed time-locked contracts (HTLC) enable routed payments to be atomic. For a technical description of HTLCs and multi-hop payments the astute reader is referred to [8].

In the following we will model LN as an undirected, weighted graph, where nodes are entities who can issue payments using payment channels which are the edges of the LN graph. The weight on the edges, capacities, are the sum of individual balances. Note, that in most cases individual balances are not known to outsiders. Only the capacity of a payment channel is public information, however one can effectively assess individual balances with handy algorithms [4].

2 Lightning Network's Topology

LN can be described as a weighted graph $G = (V, E)$, where V is the set of LN nodes and E is the set of bidirectional payment channels between these nodes. We took a snapshot[2] of LN's topology on the 10th birthday of Bitcoin, 2019 January 3rd. In the following we are going to analyze this dataset. Although the number of nodes and payment channels are permanently changing in LN, we observed through several snapshots that the main topological characteristics (density, average degree, transitivity, nature of degree distribution) remained unchanged. We leave it for future work to analyze the dynamic properties of LN.

LN gradually increased adoption and attraction throughout 2018, which resulted in 3 independent client implementations (c-lightning,[3] eclair[4] and lnd[5]) and 2344 nodes joining LN as of 2019, January 3rd. The density of a graph is defined as $D = \frac{2|E|}{|V||V-1|}$ which is the ratio of present and potential edges. As it is shown in Fig. 1. LN is quite a sparse graph. This is further justified by the fact that LN has 530 bridges, edges which deletion increases the number of connected components. Although LN consists of 2 components, the second component has only 3 nodes. The

[2] https://graph.lndexplorer.com.

[3] https://github.com/ElementsProject/lightning/.

[4] https://github.com/ACINQ/eclair.

[5] https://github.com/lightningnetwork/lnd.

Number of nodes	2344
Number of payment channels	16617
Average degree	7.0891
Connected components	2
Density	0.00605
Total BTC held in LN	543.61855₿
s-metric	0.6878
Maximal independent set	1564
Bridges	530
Diameter	6
Radius	3
Mean shortest path	2.80623
Transitivity	0.1046
Average clustering coefficient	0.304
Degree assortativity	−0.2690

Fig. 1 LN at a glance: basic properties of the LN graph

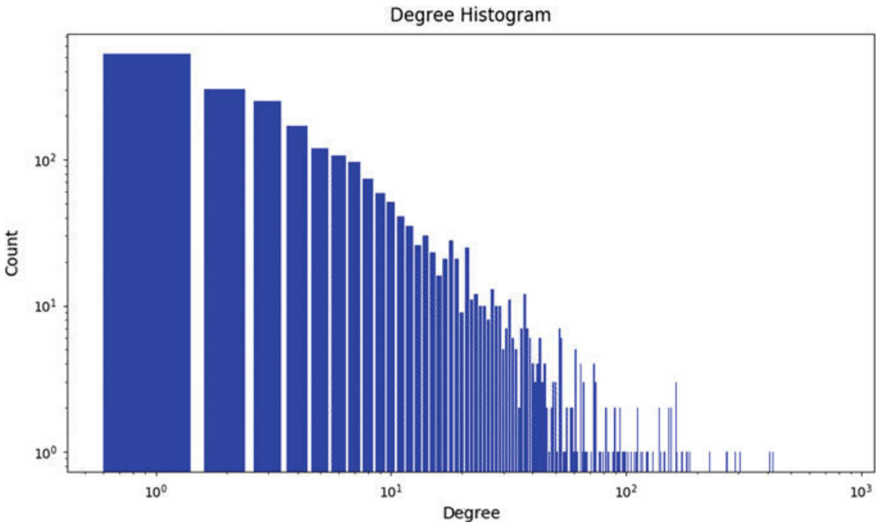

Fig. 2 LN's degree distribution

low transitivity, fraction of present and possible triangles in the graph, highlights the sparseness of LN as well.

Negative degree assortativity of the graph indicates that on average low degree nodes tend to connect to high degree nodes rather than low degree nodes. Such a dissortative property hints a hub and spoke network structure, which is also reflected in the degree distribution, see Fig. 2.

Average shortest path length is 2.80623, without taking into account capacities of edges, which signals that payments can easily be routed with a few hops throughout the whole network. Although this is far from being a straightforward task, since one also needs to take into consideration the capacity of individual payment channels along a candidate path.

When a new node joins LN, it needs to select which other nodes it is trying to connect to. In the *lnd* LN implementation key goals for a node is to optimize its centrality by connecting to central nodes. This phenomena sets up a preferential attachment pattern. Other LN implementations rely on their users to create channels manually, which also most likely leads to users connecting to high-degree nodes. Betweenness centrality of a node v is given by the expression $g(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$, where σ_{st} is the total number of shortest paths between node s and t, whilst $\sigma_{st}(v)$ is the number of those paths, that pass through v. Closeness centrality of a node v is defined as $CC(u) = \frac{N}{\sum_{u \neq v} d(u,v)}$, where N is the number of nodes in the graph and $d(u, v)$ is the distance between node u and v. Closeness centrality measures how close a node is to all other nodes.

Small-world architectures, like LN, exhibit high clustering with short path lenghts. The appropriate graph theoretic tool to asses clustering is the clustering coefficient [11]. Local clustering coefficient measures how well a node's neighbors are connected to each other, namely how close they are to being a clique. If a node u has $deg(u)$ neighbors, then between these $deg(u)$ neighbors could be at maximum $\frac{1}{2}deg(u)(deg(u) - 1)$ edges. If $N(u)$ denotes the set of u's neighbors, then the local clustering coefficient is defined as $C(u) = \frac{2|(v,w):v,w \in N(u) \wedge (v,w) \in E|}{deg(u)(deg(u)-1)}$.

LN's local clustering coefficient distribution suggestively captures that LN is essentially comprised of a small central clique and a loosely connected periphery.

2.1 Analysis of LN's Degree Distribution

LN might exhibit scale-free properties as the s-metric suggests. S-metric was first introduced by Lun Li et al. in [5] and defined as $s(G) = \sum_{(u,v) \in E} deg(u)deg(v)$. The closer to 1 s-metric of G is, the more scale-free the network. Diameter and radius of LN suggest that LN is a small world. Somewhat scale-freeness is also exhibited in the degree distribution of LN. Majority of nodes have very few payment channels, although there are a few hubs who have significantly more connections as it can be seen in Fig. 2. The scale-freeness of LN is further justified also by applying the method introduced in [2]. The maximum-likelihood fitting asserted that the best fit for the empirical degree distribution is a power law function with exponent $\gamma = -2.1387$. The goodness-of-fit of the scale-free model was ascertained by the Kolmogorov-Smirnov statistic. We found that the p-value of the scale-free hypothesis is $p = 0.8172$, which is accurate within 0.01. Therefore the scale-free hypothesis is plausible for the empirical degree distribution of LN (Fig. 3).

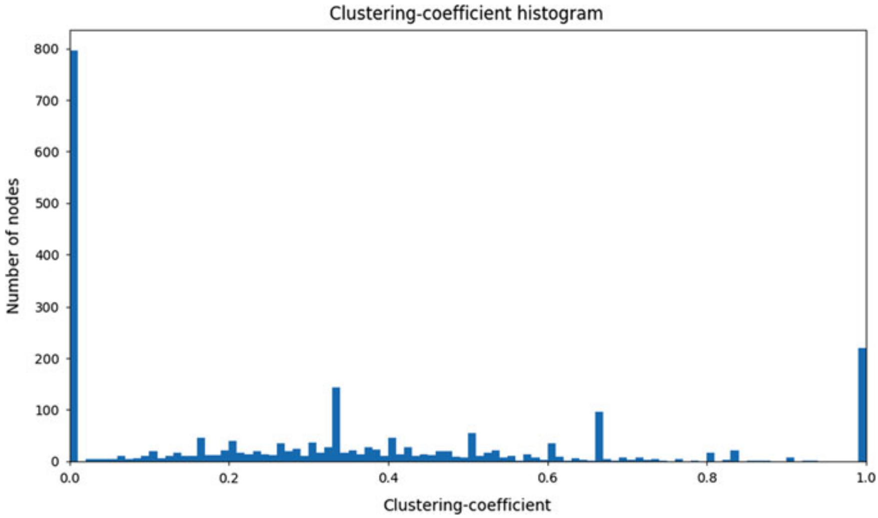

Fig. 3 Local clustering coefficient of LN

3 Robustness of LN

It is a major question in network science how robust a given network is. LN, just like Bitcoin, is a permissionless network, where nodes can join and leave arbitrarily at any point in time. Nodes can also create new payment channels or close them any time. Furthermore as new payments are made, capacities of payment channels are changing steadily. Despite the dynamic nature of LN, its topology's characteristics remain constant after all. In this section we investigate how resilient LN is, whether it can effectively withhold random node failures or deliberate attacks.

Measuring robustness means that one gradually removes nodes and/or edges from the graph, until the giant component is broken into several isolated components. The fraction of nodes that need to be removed from a network to break it into multiple isolated connected components is called the percolation threshold and is denoted as f_c. In real networks percolation threshold can be well estimated by the point where the giant component first drops below 1% of its original size [1].

3.1 Random Failures

Random failures are a realistic attack vector for LN. If nodes happen to be off-line due to bad connections or other reasons, they can not participate in routing payments anymore. Such a failure can be modeled as if a node and its edges are removed from the graph.

Fig. 4 Random failures in networks. Values of critical thresholds for other real networks are taken from [1]

Network	f_c
Internet	0.92
WWW	0.88
US Power Grid	0.61
Mobil Phone Call	0.78
Science collaboration	0.92
E. Coli Metabolism	0.96
Yeast Protein Interactions	0.88
LN	0.96

For scale-free networks with degree distribution $P_k = k^{-\gamma}$, where $2 < \gamma < 3$ the percolation threshold can be calculated by applying the Molloy-Reed criteria, ie. $f_c = 1 - \frac{1}{\frac{\gamma-2}{3-\gamma} k_{min}^{\gamma-2} k_{max}^{3-\gamma} - 1}$, where k_{min} and k_{max} denote the lowest and highest degree nodes respectively. This formula yields $f_c = 0.9797$ for LN in case of random failures. This value is indeed close to the percolation threshold measured by network simulation as shown in Fig. 4, that is, LN provides an evidence of topological stability under random failures. In particular this is due to the fact that in LN a randomly selected node is unlikely to affect the network as a whole, since an average node is not significantly contributing to the network's topological connectivity, see also degree distribution at Fig. 2.

3.2 Targeted Attacks

Targeted attacks on LN nodes are also a major concern as the short history of LN has already shown it. On 2018 March 21st,[6] 20% of nodes were down due to a Distributed Denial of Service (DDoS) attack against LN nodes. Denial of Service (DoS) attacks are also quite probable by flooding HTLCs. These attack vectors are extremely harmful, especially if they are coordinated well. One might expect that not only state-sponsored attackers will have the resources to attack a small network like LN. In the first attack scenario we removed 30 highest-degree nodes one by one starting with the most well-connected one and gradually withdraw the subsequent high-degree nodes. We recorded the number of connected components. As it is shown in Fig. 5. even just removing the highest-degree node[7] fragments the LN graph into 37 connected components! Altogether the removal of the 30 largest hubs incurs LN to collapse into 424 components, although most of these are isolated vertices. This symptom can be explained by the experienced dissortativity, namely hubs tend to be at the periphery.

We reasserted the targeted attack scenario, but for the second time we only removed one of the 30 largest hubs and recorded the number of connected com-

[6] https://www.trustnodes.com/2018/03/21/lightning-network-ddos-sends-20-nodes.
[7] http://rompert.com/.

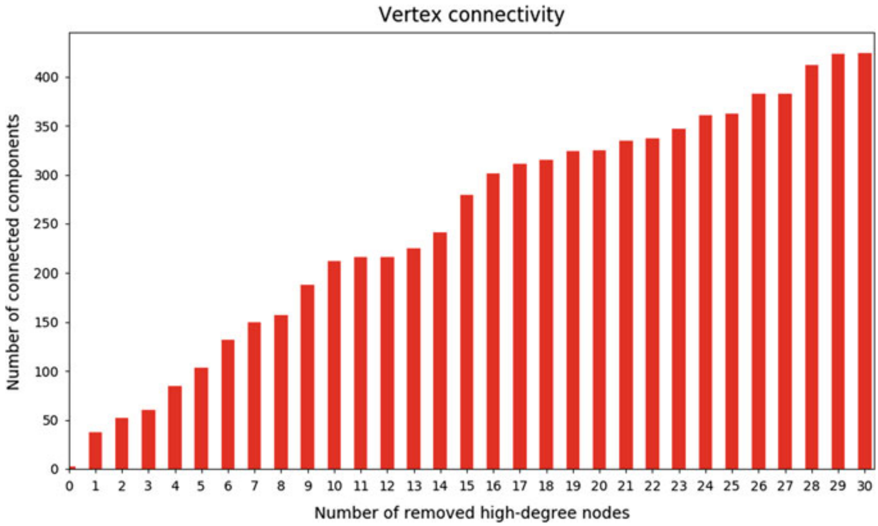

Fig. 5 LN's vertex connectivity, when all the 30 largest hubs are removed one by one

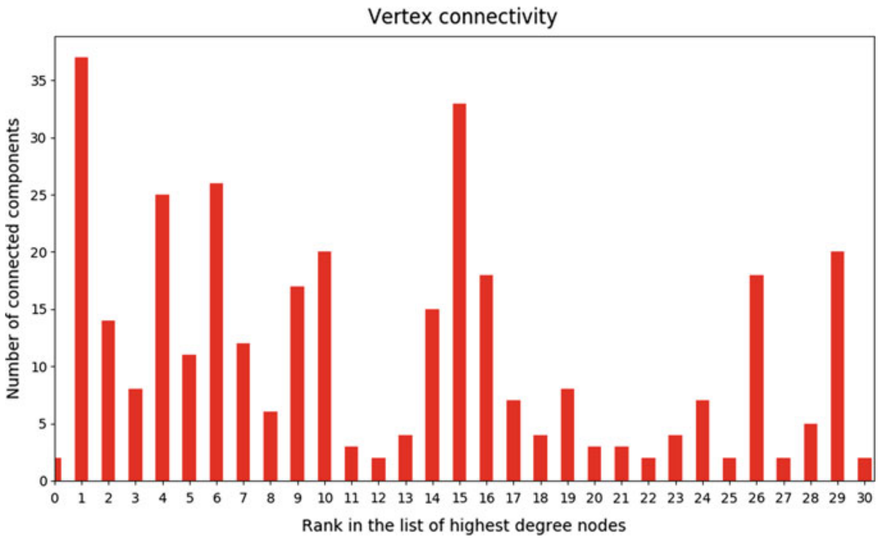

Fig. 6 LN's vertex connectivity if only one high-degree node is removed from the graph

ponents. As it can be seen in Fig. 6 most of the hubs, 25, would leave behind several disconnected components (Fig. 7).

Such network fragmentations are unwanted in case of LN, because they would make payment routing substantially more challenging (one needs to split the payment over several routes) or even impossible (there would be no routes at all).

Fig. 7 Real networks under targeted attacks. Values of critical thresholds for other real networks are taken from [1] to [6]

Network	f_c
Internet	0.16
WWW	0.12
Facebook	0.28
Euroroad	0.59
US Power Grid	0.20
Mobil Phone Call	0.20
Science collaboration	0.27
E. Coli Metabolism	0.49
Yeast Protein Interactions	0.16
LN	0.14

Furthermore we estimated the percolation threshold by simulating two attacking strategies. In the first scenario we removed high degree nodes one by one (high degree removal attack, HDR) and in the second we removed nodes with the highest betweenness centrality (high betweenness removal attack, HBR). Note that in both cases after each node removal we recalculated which node has the highest degree or betweenness centrality in order to have a more powerful attack. We found out that $f_c = 0.1627$ for removing high degree nodes, while for removing high betweenness centrality nodes $f_c = 0.1409$, therefore choosing to remove high betweenness centrality nodes is a better strategy as it can also be seen in Fig. 10.

Node outage not only affects robustness and connectivity properties, but also affects average shortest path lengths and available liquidity. Although the outage of random nodes does not significantly increase the average shortest path lengths in LN, targeted attacks against hubs increase distances between remaining nodes. The spillage of high-degree nodes not only decreases the amount of available liquidity but also rapidly increases the necessary hops along payment routes as Figs. 8 and 9 suggest. This could cause increased ratio of failed payments due to larger payment routes and sparser liquidity. Figure 9 demonstrates how capital allocation is centred upon a few high degree nodes, namely already the removal of as few nodes as 37 decreases the available liquidity by more than 50%. Unfortunately most of these nodes are run by a handful of companies (Fig. 10).

3.3 *Improving LN's Resilience Against Random Failures and Attacks*

Designing networks which are robust to random failures and targeted attacks appear to be a conflicting desire [1]. For instance a star graph, the simplest hub and spoke network, is resilient to random failures. The removal of any set of spokes does not hurt the connectedness of the main component. However it can not withstand a targeted attack against its central node, since it would leave behind isolated spokes. Furthermore when one attempts increasing robustness of a network, they do desire not to decrease the connectivity of nodes.

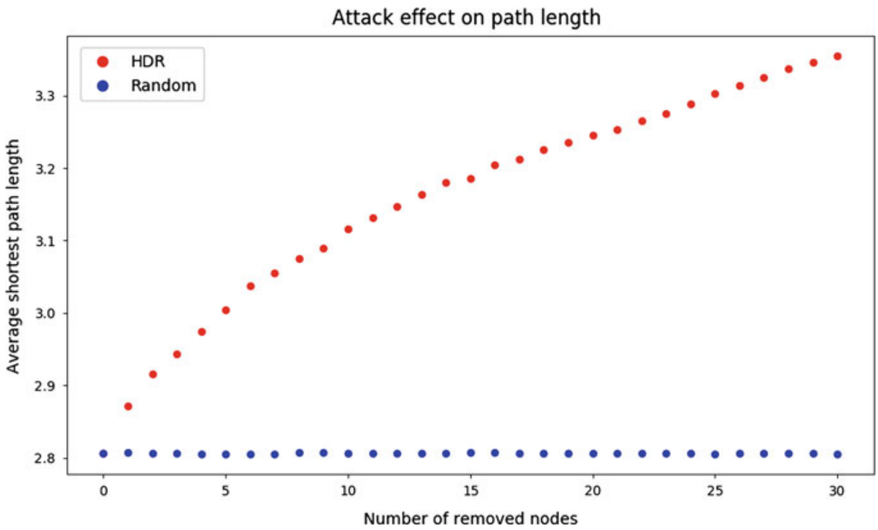

Fig. 8 High degree removal (HDR) attack effects average shortest path lengths

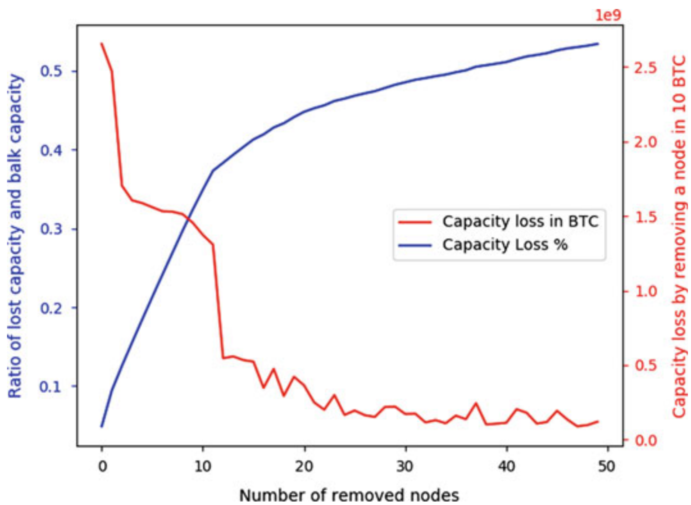

Fig. 9 Lost capacity as removing high-degree nodes

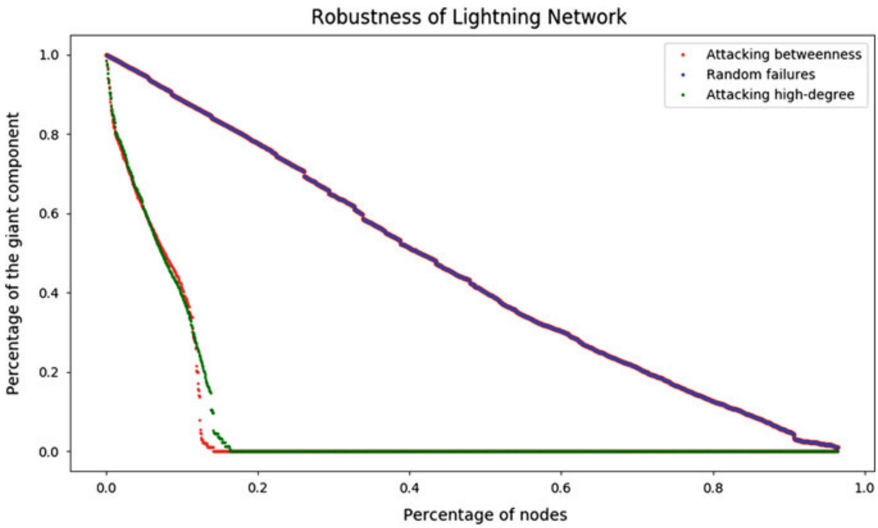

Fig. 10 Percolation thresholds for various attack scenarios: $f_c^{HDR} = 0.1627$, $f_c^{HBR} = 0.1409$, $f_c^{RND} = 0.9645$

A similar optimization strategy of robustness and connectivity to that of [10] could be applied to LN as well. We leave it for future work to empirically assess the robustness and connectivity gains if the strategy of [10] would be implemented in LN client implementations.

Nonetheless, we can still enhance the network's attack tolerance by connecting its peripheral nodes [1]. This could be achieved by LN client implementations by implicitly mandating newcomers to connect to not only hubs, as current implementations do, but also to at least a few random nodes.

4 Conclusion

In summary, a better understanding of the network topology is essential for improving the robustness of complex systems, like LN. Networks' resilience depends on their topology. LN is well approximated by the scale-free model and also its attack tolerance properties are similar to those of scale-free networks; in particular, while LN is robust against random failures, it is quite vulnerable in the face of targeted attacks. High-level depictions of LN's topology convey a false image of security and robustness. As we have demonstrated, LN is structurally weak against rational adversaries. Thus, to provide robust Layer-2 solutions for blockchains, such as LN and Raiden, the community needs to aim at building resilient network topologies.

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References

1. Barabási, A.L., et al.: Network Science. Cambridge University Press (2016)
2. Clauset, A., Shalizi, C.R., Newman, M.E.: Power-law distributions in empirical data. SIAM Rev. **51**(4), 661–703 (2009)
3. Gudgeon, L., Moreno-Sanchez, P., Roos, S., McCorry, P., Gervais, A.: Sok: Off the chain transactions. Cryptology ePrint Archive, Report 2019/360 (2019). https://eprint.iacr.org/2019/360
4. Herrera-Joancomarti, J., Navarro-Arribas, G., Ranchal-Pedrosa, A., Pérez-Sola, C., Garcia-Alfaro, J.: On the difficulty of hiding the balance of lightning network channels. Cryptology ePrint Archive, Report 2019/328 (2019). https://eprint.iacr.org/2019/328
5. Li, L., Alderson, D., Doyle, J.C., Willinger, W.: Towards a theory of scale-free graphs: definition, properties, and implications. Internet Math. **2**(4), 431–523 (2005)
6. Lin, Y., Chen, W., Zhang, Z.: Assessing percolation threshold based on high-order non-backtracking matrices. In: Proceedings of the 26th International Conference on World Wide Web, International World Wide Web Conferences Steering Committee, pp. 223–232 (2017)
7. Nakamoto, S.: Bitcoin: A Peer-to-Peer Electronic Cash System (2008)
8. Poon, J., Dryja, T.: The bitcoin lightning network: scalable off-chain instant payments (2016). https://lightning.network/lightning-network-paper.pdf
9. Roos, S., Moreno-Sanchez, P., Kate, A., Goldberg, I.: Settling payments fast and private: efficient decentralized routing for path-based transactions (2017). arXiv preprint arXiv:1709.05748
10. Shargel, B., Sayama, H., Epstein, I.R., Bar-Yam, Y.: Optimization of robustness and connectivity in complex networks. Phys. Rev. Lett. **90**(6), 068701 (2003)
11. Watts, D.J., Strogatz, S.H.: Collective dynamics of 'small-world' networks. Nature **393**(6684), 440 (1998)

Ping-Pong Governance: Token Locking for Enabling Blockchain Self-governance

Paul Merrill, Thomas H. Austin, Justin Rietz and Jon Pearce

Abstract Updating blockchain-based protocols remains a significant challenge. If the community does not come to an agreement, a *hard-fork* can occur, splitting the blockchain's community. Previous protocols have provided mechanisms to establish community consensus through the protocol itself, but these protocols either facilitate substantial, infrequent updates, or they allow more frequent but only minor changes. This work offers a mechanism that allows clients to vote by locking tokens, making the clients' tokens temporarily unavailable in exchange for their vote. This design introduces an economic cost to voting, allowing us to measure both breadth and depth of support. Since there is an economic cost to voting, we wish to make non-contentious issues cheap to pass, but still allow the community to establish agreement on larger, more disputatious proposals. We achieve this property by a *ping-pong governance model*. An issue is tentatively accepted when it achieves enough votes within a fixed period. The proposal then enters a review period, where the opponents must gather enough votes to veto it. The supporters then have their own opportunity to overrule the veto. This process continues with new voting rounds until one side is unable to exceed the needed threshold, settling the issue. Our simulations show that this model allows the community to come to agreement quickly on popular changes, but still come to resolution when the community is more divided. Finally, we define the ideal properties of a blockchain governance protocol, and evaluate different governance protocols under these criteria.

Keywords Cryptocurrencies · Governance · Token economics

P. Merrill · T. H. Austin
0Chain LLC, Cupertino, USA

P. Merrill · T. H. Austin (✉) · J. Rietz · J. Pearce
San José State University, San José, USA
e-mail: thomas.austin@sjsu.edu

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1 Introduction

Bitcoin [14] introduced the blockchain and revolutionized distributed applications. A tremendous number of innovations have been built on this concept: Ethereum [21] introduced a Turing-complete programming language on the blockchain; Blockstack [1] showed how it could be used to build a public-key infrastructure; several others [5, 13, 19, 20] leveraged the blockchain to provide distributed storage.

But blockchain-based protocols are difficult to update once launched. Upgrading requires participants to explicitly opt-in to the new protocol by installing new versions of the software. These *hard forks* are undesirable, since they can require significant manual involvement. Furthermore, in cases where the blockchain's community fails to come to an agreement, it can split into two blockchains; the infamous DAO debacle [10] that split Ethereum and Ethereum Classic is perhaps the best known of these.

Governance in the context of blockchain refers to how a blockchain's community can make changes that involve some amount of human interaction [4]. While governance protocols cannot guarantee that the blockchain will not fork over a particularly contentious issue like the DAO debacle, they can provide an authoritative community decision that automates much of the process in order to minimize the work requiring human intervention. We discuss the desired properties of governance protocols in Sect. 2.

Other protocols have attempted to offer a smoother process for the protocol to evolve. Tezos [6] offers one extreme, where an entirely new code base can be proposed. However, due to the potential significance of changes, they can happen only infrequently, and require substantial involvement from the community. Other protocols such as Cypherium [7] offer more limited, but more frequent changes. In the case of Cypherium specifically, changes are limited to minor, 1% configuration adjustments.

In this paper, we seek to provide a voting mechanism that can allow minor and popular changes to happen quickly, but that will also allow more substantial (and potentially controversial) changes.

Our design builds on the token-locking reward model developed by Merrill et al. [12]. In this model, clients who own tokens may temporarily lock them to produce token rewards for service providers. Clients may instead choose to spend their tokens, which produces greater rewards for fewer tokens. However, clients who choose this option lose the ability to re-use their tokens, giving up the "free" aspect of the token-locking reward model.

We leverage this token-locking model to serve as a mechanism for the community to vote on changes. Token rewards are treated as votes; by locking more tokens, clients may dedicate more votes to a proposal that they favor. Similarly, they may allocate token rewards against any proposals that they oppose. A crucial aspect of our design is that our voting mechanism can measure both whether a proposal has broad community support and the degree of support or opposition from different parties for a specific proposal. In extreme cases, smaller stake holders can choose to spend their tokens to increase their voting power. In this manner, a group of clients with a relatively small amount of stake (but with strong support for an initiative) have some hope of defeating opponents with less stake but who are not as strongly motivated.

One concern in voting protocols is that a wealthy supporter of a proposal could sweep in during the last moment of voting to pass a measure before other members of the community could react. This is a particular concern for our protocol, since clients have an economic cost associated with voting for a proposal. Our design addresses this issue by having multiple rounds of voting. If a proposal passes, it is followed by a review period where the community may veto the proposal. If a proposal if vetoed, the community may vote to override the veto. The override itself may be vetoed, which may also be overridden, and so on. To refer to vetoing and overriding generically, we say that a vote is *overturned*. Eventually, one side or the other will exceed a threshold that the other side cannot match and the issue will be settled. To minimize the back and forth votes, the overturning faction must exceed the threshold by a fixed buffer amount.

We refer to our model as the *ping-pong governance model*. Each side must gather enough votes to exceed the threshold to overturn a decision, akin to hitting a ping-pong ball over the net. In ping-pong, the game ends when one side fails to get the ball over the net; in our protocol, the voting ends when one side fails to gather sufficient votes to exceed the overturn threshold.

This paper is organized as follows: Sect. 2 defines the ideal properties of a blockchain governance protocol; Sect. 3 reviews the design of the token-locking reward model, which is the mechanism powering our voting model; Sect. 4 reviews the voting mechanism and the types of changes that we wish to allow; Sect. 5 walks through an example of a proposal going through several rounds of voting; Sect. 6 presents our experimental evidence; Sect. 7 discusses our design; Sect. 8 reviews other blockchain governance protocols; and Sect. 9 concludes.

2 Ideal Properties of a Blockchain Governance Protocol

Different governance protocols for blockchains have different priorities. In this section, we define several properties that would be ideal for a blockchain governance protocol. An ideal governance protocol should offer:

1. *Transparency*. All members of the blockchain community should be able to observe the governance process.
2. *Fairness*. What is "fair" depends on the intentions of the blockchain community. In most, voting power should match the stake of the voter.
3. *Automatic enforceability*. Once a decision has been made, all clients on the blockchain should transition to the new model automatically. In other words, the decision of the governance protocol should be enforced automatically.

4. *Flexibility*. A protocol should be able to handle minor tweaks as well as substantial revisions to the blockchain protocol.
5. *Timeliness*. Especially in the case of urgent issues, the governance protocol should be able to come to consensus relatively quickly.

Our own protocol attempts to be transparent, fair, and flexible. We also seek automatic enforceability and timeliness for more minor changes to the protocol. In terms of fairness, we strive to find a balance between the interests of the major stakeholders and the members of our community at large. We believe that introducing an economic cost to voting is an elegant way of balancing the concerns; major stakeholders are more likely to win on most votes, but are unlikely to override the community if the community at large is more passionate about an issue (and willing to suffer a greater economic cost, relative to their total wealth).

Other protocols have different priorities. Tezos [6] offers all of these properties except for timeliness; changes in their ecosystem are deliberately slow, in order to prevent negative changes from happening too quickly. Cypherium [7] prioritizes timeliness and automatic enforceability, at a cost to fairness (only the validator nodes may vote) and flexibility (only minor configuration changes are permitted).

In Sect. 8, we review several blockchain governance protocols and discuss how they align with these properties.

3 Review of Token Locking Reward Model

We first review the token-locking reward model [12], since that is integral to the design of our voting mechanism. The token-locking reward model was designed to allow clients with tokens to write transactions for "free", in the sense that the clients do not permanently lose their tokens.

For comparison, Bitcoin [14] pays miners with two distinct mechanisms. Coinbase transactions create new tokens to reward miners for creating blocks, regardless of what transactions they produce. The new tokens increase the supply of available bitcoins, meaning that the network is paying miners through inflation. Transaction fees are a second mechanism for rewarding miners in the Bitcoin protocol. Each client who wants their transaction included in a block specifies some amount of bitcoin as a reward to the miner for including the transaction. This mechanism also serves to ease congestion, since clients may specify a larger transaction fee if they are more eager for their transaction to be included.

The token-locking reward model combines these two mechanisms. A client *locks* some tokens for a set period (usually 90 days). In exchange, new tokens are immediately created that can be used to offer a reward to miners for including a transaction in a block.

An advantage of this design is that these token rewards can be used to pay a variety of other service providers. For instance, Merrill et al. [12] allow clients to lock tokens to pay storage providers, referred to as *blobbers*. This design stands in contrast to other protocols such as Tendermint [9], where miners can lock tokens to produce blocks and gain token rewards, but where the mechanism is not designed to pay any other parties.

In this sense, voting can be thought of as an additional service. By locking tokens, clients can generate “rewards” of votes for a proposal. This design introduces an economic cost to voting, which therefore serves as a mechanism for measuring a client’s commitment to an issue. In other words, we expect that disinterested clients are likely to lock very few tokens, whereas interested clients will lock more. Section 4 reviews this design in greater detail.

3.1 Transitioning Between Free and Pay Model

In some cases, clients might wish to offer greater rewards than they are able to generate by locking tokens alone. Therefore, the token-locking reward model also allows clients to spend tokens as well.

To facilitate a smooth transition between free and pay models, a client can specify some combination of tokens to be locked and of tokens to be given to the service provider. Since token rewards are paid immediately, the service provider’s calculation remains simple. For instance, a miner will select the transactions that offer the highest reward, regardless of whether the rewards were generated by locking tokens or simply given as a reward.

The total reward (R) offered to miners is determined by the interest rate multiplier (M) and the amount of tokens that the client offers to lock (A_{lock}). Additionally, a client may offer to give additional tokens (A_{spend}) to the miner. Giving tokens may be useful when the client wishes to offer a more substantial incentive than they could do by locking tokens alone. The formula for the miner’s total reward is given by the formula below:

$$R = (M * A_{lock}) + A_{spend}$$

After the locking period has ended, the client regains the tokens they locked and can use them again, but their spent tokens are gone forever.

3.2 Generating Interest and Staking Tokens

One issue with this reward model is that clients might create “faux services” as a way of generating interest for themselves. To avoid this problem, the reward model legitimizes this behavior—clients may openly lock tokens on the blockchain solely to pay themselves interest.

Service providers may be required to *stake* tokens as well; that is, they might lock tokens which may be sacrificed if they fail to perform their duties. This could make offering a service less attractive, if the service provider was forced to sacrifice the